Time Delay and Counter

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Introduction to Counter and Time Delay

- Counters are used primarily to keep track of events.

OR

A counter is a loop that counts the number of times an operation is repeated.

- Time delay is used to set timing between two events.

Time Delay Using One Register

- Code for Delay Using One Register (C):
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```
        MVI C, FFH     ; Load 255 into register C
DELAY: DCR C           ; Decrement register C
       JNZ DELAY ; Jump to DELAY if C ≠ 0
       HLT       ; Halt after delay is over
```

Timing Explanation

- Instruction Timing (T-states):
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  - MVI C, FFH = 7 T-states (once)
  - DCR C = 4 T-states (255 times)
  - JNZ DELAY =
    - 10 T-states (254 times, jump)
    - 7 T-states (1 time, no jump)

- Total T-states ≈ 3574
  Clock = 3.072 MHz → Clock Period ≈ 0.326 µs
  Delay ≈ 3574 × 0.326 µs ≈ 1.16 milliseconds

Time Delay Using Nested Loop

- Nested Loop Delay Example (Using B and C Registers):
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```
MVI B, FFH        ; Load outer loop (255)
OUTER: MVI C, FFH ; Load inner loop (255)
INNER: DCR C
       JNZ INNER
       DCR B
       JNZ OUTER
```

Time Delay Calculation

- Calculate Total Time Delay:
-

  Assume: Clock = 3.072 MHz → T = 0.326 µs

  Instructions:
  - MVI B, FFH = 7 T-states (once)
  - MVI C, FFH = 7 T-states × 255 = 1785
  - DCR C + JNZ = 14 Tx255x255=910350
- - DCR B + JNZ = 14Tx255=3570
  Total T-states ≈ 7+ 1785 +910350+3570 = 915712
  Delay ≈ 915712 × 0.326 µs ≈298.522112ms
- 2.59 ms

Time Delay Using Register pair

Instruction	T-states	Occurrences	Total
LXI D, FFFFH	10	1	10
DCX D	6	65,535	393,210
MOV A, D	4	65,535	262,140
ORA E	4	65,535	262,140
JNZ DELAY	10 × 65,534 + 7	65,535	655,347
HLT	7	1	7

Total ≈ 1,572,854 T-states

 Delay Time ≈

`1,572,854 × 0.326 µs ≈ 512.7 milliseconds`

Application based Numerical

WAP to count continuously in hexadecimal from FFH to 00H in a system with 0.5 µs clock period. Use register C to set up 1 ms delay between each count and display the number at one of the output ports.

Instructions with T-states:

```
      MVI B, 00H       ; Initialize B
Next: DCR B      ; 4T
       MVI C, Count ; 7T
Delay: DCR C     ; 4T
      JNZ Delay  ; 10T
      MOV A, B   ; 4T
      OUT PORT1  ; 10T
      JMP Next   ; 10T
```

Delay Calculation:

Loop Instructions: DCR C + JNZ Delay = 14T Delay outside loop = 4T + 10T + 10T + 7T + 4T = 35T

Given T = 0.5 µs, and desired delay = 1 ms

Delay = (14T × Count) + 35T

1 ms = (14 × Count × 0.5 µs) + (35 × 0.5 µs)

→ 1000 µs = (7 × Count) + 17.5 µs

→ Count ≈ (1000 - 17.5) / 7 = 140 (decimal) = 8C H

Problem: Zero to Nine Counter

- WAP to count from "0" to "9" with one second of delay between each count. At the count of 9, the counter should reset itself to "0" and repeat the counter sequence continuously. Use register-pair H-L to set up delay and display each count at one of the output ports. Assume the clock frequency of the microprocessor is 1 MHz.

Program

```
Next:   MVI B, 00H      ; Initialize B with 0
        MOV A, B        ; Move count to A
Display:OUT PORT1       ; Output count to port
        LXI H, Count    ; Load HL with delay count (10T)
loop:   DCX H           ; Decrement HL (6T)
        MOV A, H        ; Move H to A (4T)
        ORA L           ; OR L with A (4T)
        JNZ loop        ; If HL ≠ 0, repeat (10T)
        INR B           ; Increment counter (4T)
        MOV A, B        ; Move B to A (4T)
        CPI 0AH         ; Compare with decimal 10 (7T)
        JNZ Display     ; If not 10, go back to display (10T)
        JMP Next        ; Reset to 0 and repeat (10T)
```

Delay Loop T-states Calculation:

The delay loop contains:

DCX H = 6T

MOV A, H = 4T

ORA L = 4T

JNZ loop = 10T

Total per iteration:

6 + 4 + 4 + 10 = 24 T-states

Total Delay

Total Delay = (Delay in loop) + (Delay outside loop)

= [24T × Count] + 45T

Clock period T = 1 μs = 10^{-6} sec

Total required delay = 1 second

1 sec = ($24 \times 10^{-6} \times$ Count) + (45×10^{-6})

Count = ($1 - 45 \times 10^{-6}$) / (24×10^{-6})

≈ 0.999955 / 0.000024

≈ 41664.79 ≈ 41665 (rounded)

41665 (decimal) = A2C1H